

The Cost of Complexity: Hardware-Induced Bottlenecks and Failure Modes in Real-Time Deep Reinforcement Learning

Abstract

Integrating high-capacity representation models, such as meta-learning context encoders and recurrent state-space models (RSSMs), into continuous-control reinforcement learning (RL) has shown strong promise in simulated environments. However, deploying these complex architectures in real-time, high-speed control tasks under strict hardware constraints remains profoundly challenging. In this paper, we analyze **SPECTRE**, an advanced Soft Actor-Critic (SAC) variant equipped with PEARL-style context adaptation, MC-dropout uncertainty estimation, and curiosity-driven exploration, applied to the continuous racing environment Trackmania. Evaluated on constrained hardware (Intel i5 CPU, GPU with 4 GB VRAM), we empirically show how compounded computational overhead and architectural complexity introduce severe latency, exacerbate catastrophic local-minima traps, and destabilize training relative to lightweight baselines. We find that VRAM saturation and inference latency can effectively violate the practical assumptions required for Markov decision process (MDP)-based control, causing decoupled state-action feedback loops. Our analysis highlights the necessity of compute-aware architectural design when transitioning modern RL beyond toy environments and toward real-world deployment.

1 Introduction

Deep reinforcement learning has achieved remarkable proficiency in continuous control. To address persistent challenges such as sample inefficiency and poor generalization across varying dynamics, modern architectures increasingly incorporate auxiliary representation models alongside core policy networks. Methods such as PEARL (Probabilistic Embeddings for Actor-Critic RL) introduce latent context encoders for rapid online adaptation, while recurrent state-space models enable world modeling, forward prediction, and epistemic uncertainty estimation.

However, the reported efficacy of these methods is predominantly established in forgiving, non-real-time simulated environments (e.g., MuJoCo or DMControl) and often with high-end compute resources. In real-world continuous-control settings, such as autonomous racing or high-speed robotics, the environment does not pause for agent inference. Under such conditions, hardware constraints (e.g., limited VRAM and CPU/GPU throughput bottlenecks) are not merely implementation inconveniences; they can fundamentally alter the learning dynamics.

This paper investigates the failure modes that arise when integrating complex representation models into DroQSAC (a dropout-regularized SAC variant) in the Trackmania racing environment under strict hardware limitations (NVIDIA GTX 1650 with 4 GB VRAM, Intel i5-12450H CPU). We implement a unified architecture, **SPECTRE**, combining a PEARL-style contextual encoder, an RSSM decoder with Monte Carlo dropout, and prediction-error intrinsic curiosity.

Although this design is theoretically capable of substantial generalization, our empirical evaluation reveals systemic instability in constrained real-time training. We identify three primary bottleneck-induced failure modes:

1. **Compute-induced MDP mismatch:** High inference latency from the combined RSSM and context encoder desynchronizes the agent’s observation-action loop from the environment’s true physical trajectory.
2. **Exacerbated local-minima capture:** Coupled gradients between high-capacity contextual modules and the actor can produce rigid, overconfident policies early in training (e.g., persistent stalling or repeated wall-crashing), limiting the intended benefits of curiosity.
3. **VRAM saturation and uncertainty collapse:** Auxiliary modules heavily consume the 4 GB memory budget, forcing batch-size and replay compromises that destabilize uncertainty estimation and downstream DroQSAC updates.

We present an ablation study comparing full SPECTRE against lightweight baselines (vanilla SAC and standard DroQSAC) to quantify the practical cost of architectural complexity and to frame capacity–latency trade-offs in real-time RL.

2 Related Work

Sample-Efficient RL and Soft Actor-Critic (SAC). Maximum-entropy RL methods such as SAC provide a robust baseline for continuous control. Recent variants, including DroQSAC, replace large critic ensembles with dropout-regularized critics to reduce overestimation while lowering parameter counts.

Meta-RL and Context Adaptation. Approaches such as PEARL use a context encoder to map recent transitions to a latent task embedding, enabling fast adaptation under shifting dynamics. In practice, this encoder introduces additional recurrent or aggregation overhead that can strain strict inference budgets.

Uncertainty Estimation and Curiosity. World-model approaches, including RSSM-style latent dynamics models, generate intrinsic rewards from prediction error or surprise. Reliable epistemic uncertainty typically relies on MC-dropout or ensembles, which increase computational load roughly linearly in the number of stochastic forward passes.

Real-Time RL and Latency. Prior work shows that action delays and asynchronous control loops significantly degrade policy quality and sample efficiency. When environment dynamics evolve continuously outside the agent’s decision latency budget, off-policy learning can accumulate substantial off-environment bias.

Drafting Note

Methodology and Results sections are planned next.